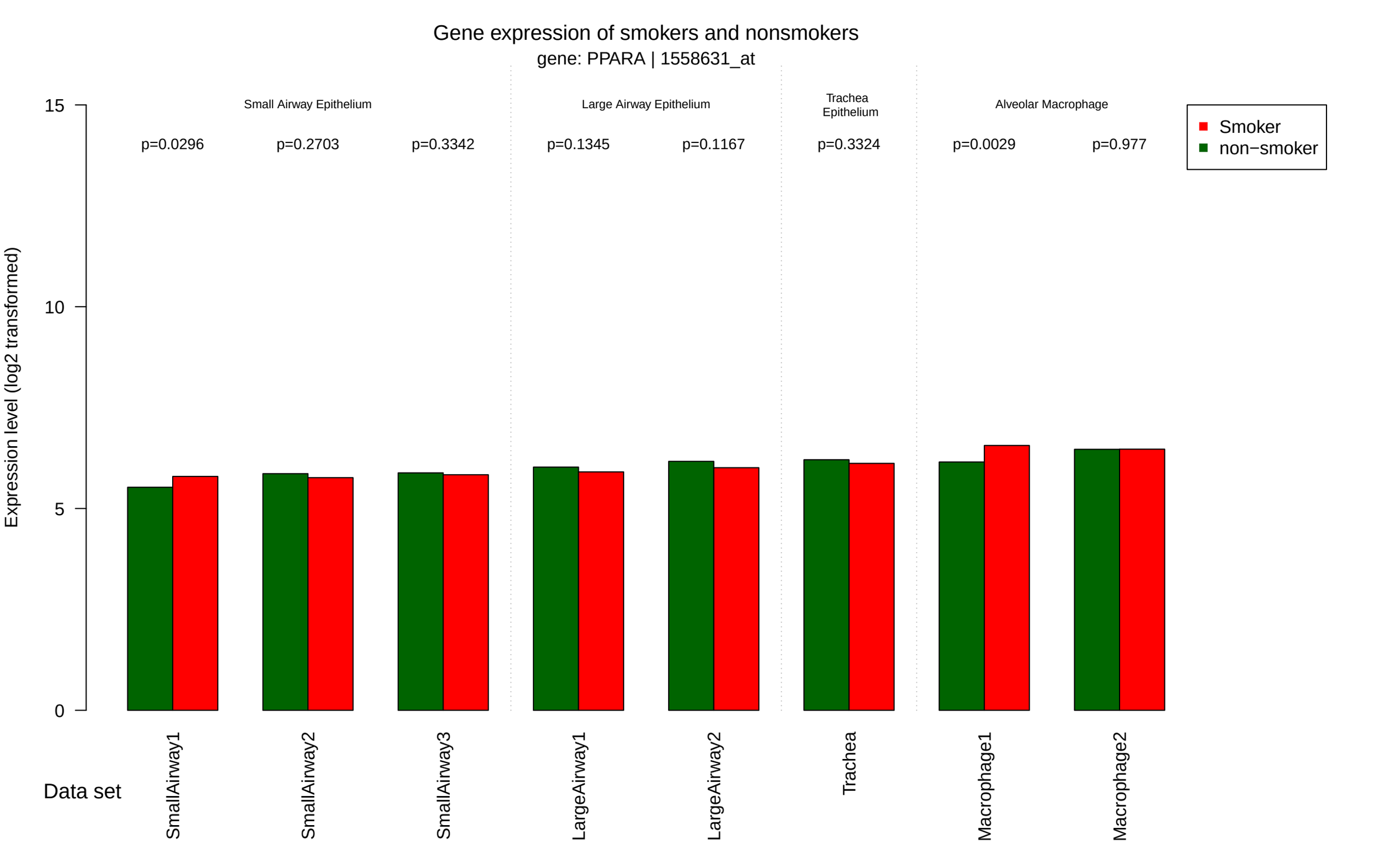
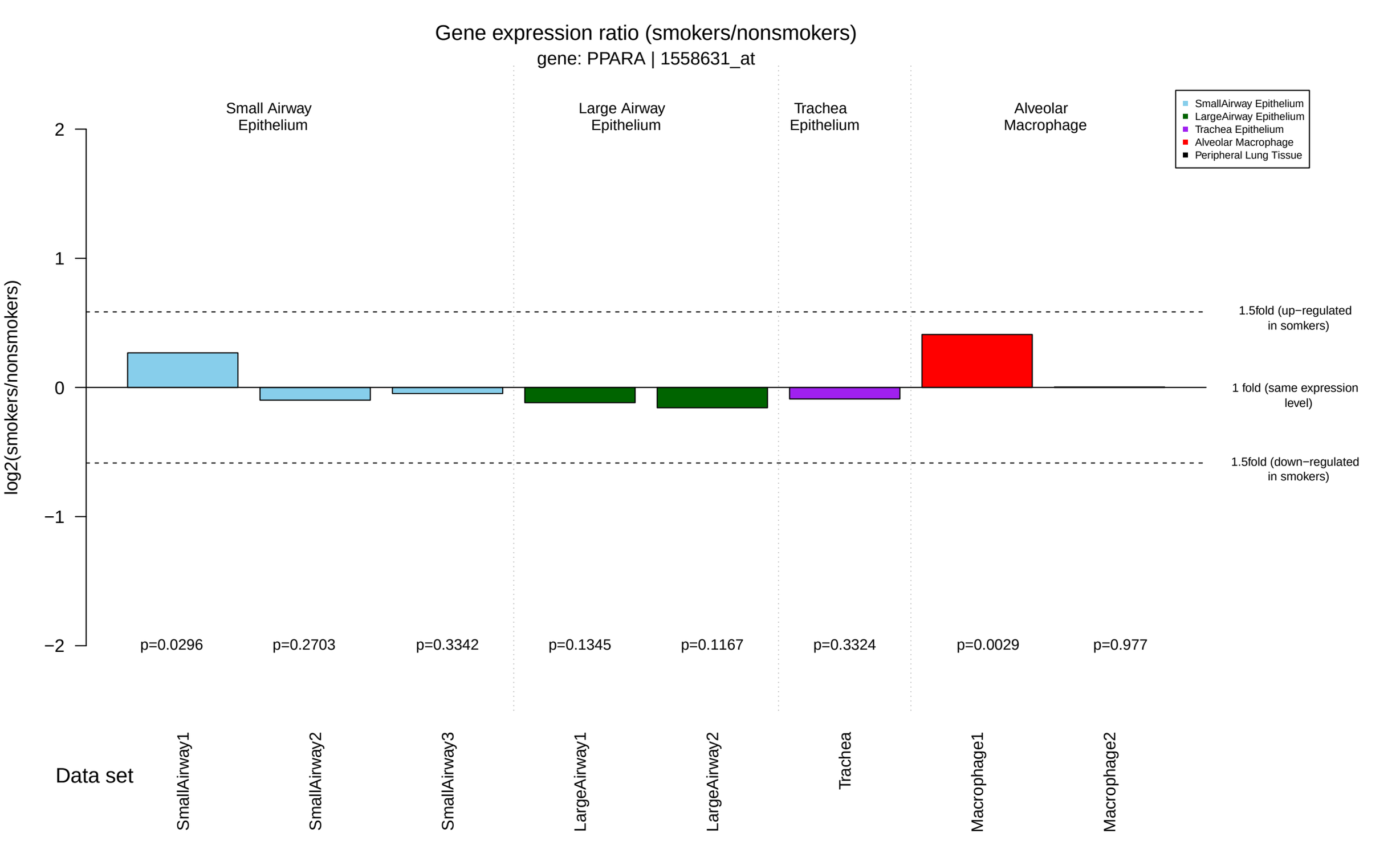
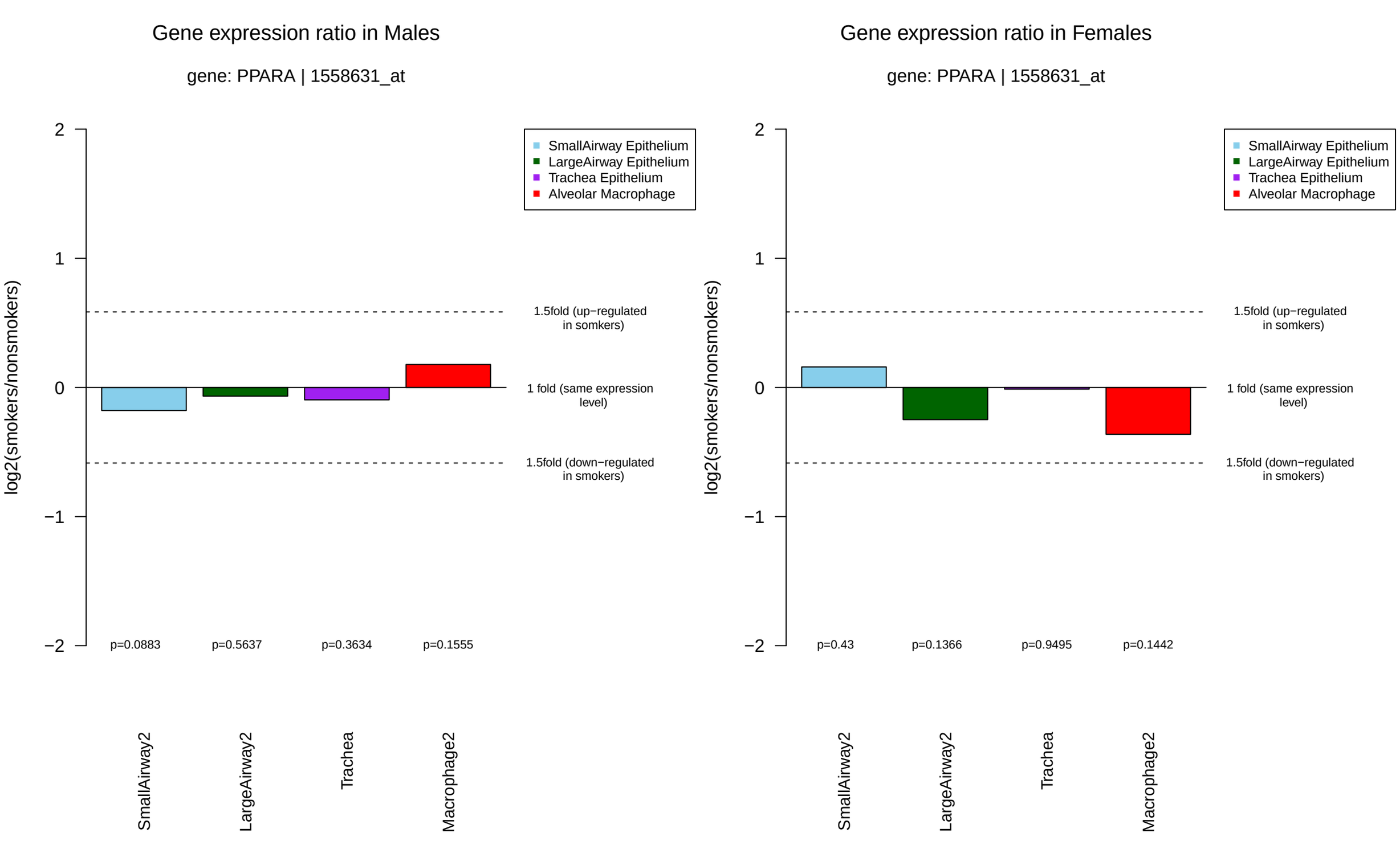
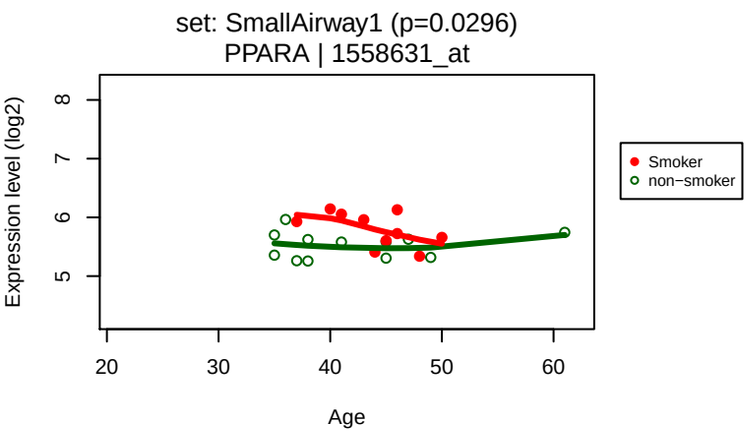
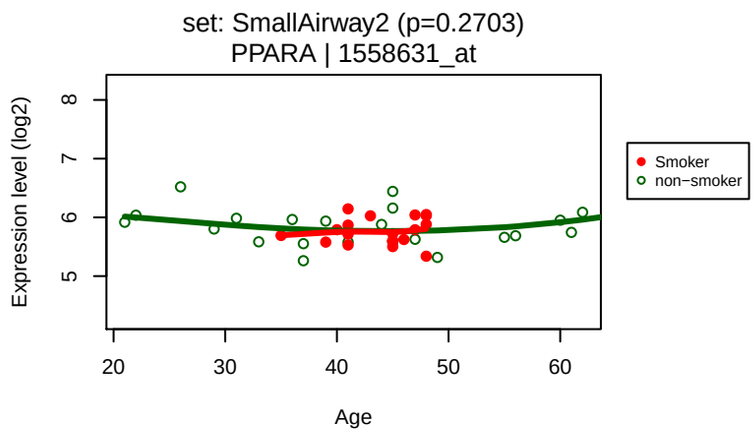
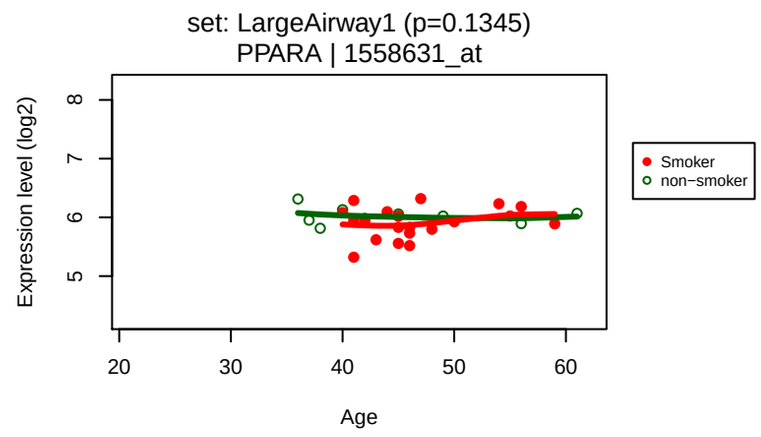

Gene expression of smokers and nonsmokers
gene: PPARA | 1558631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0296
p=0.2703
p=0.3342
p=0.1345
p=0.1167
p=0.3324
p=0.0029
p=0.977
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

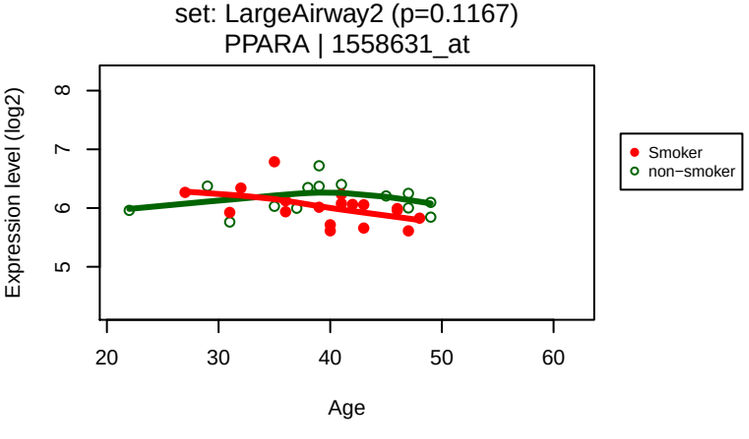

Gene expression ratio (smokers/nonsmokers)
gene: PPARA | 1558631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0296
p=0.2703
p=0.3342
p=0.1345
p=0.1167
p=0.3324
p=0.0029
p=0.977
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

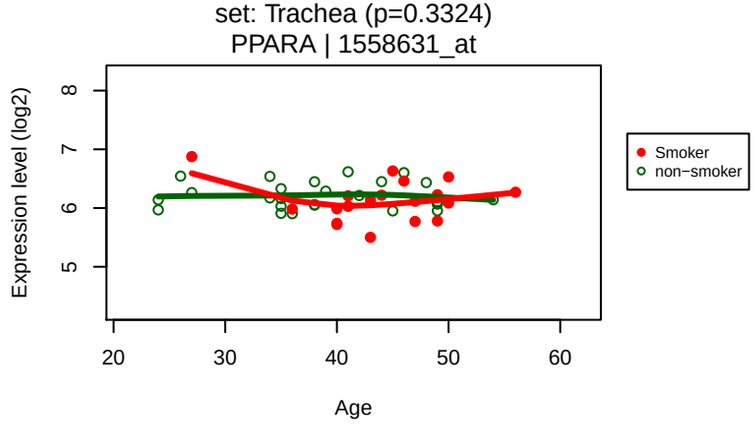

Gene expression ratio in Males
gene: PPARA | 1558631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0883
p=0.5637
p=0.3634
p=0.1555
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPARA | 1558631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.43
p=0.1366
p=0.9495
p=0.1442
SmallAirway2
LargeAirway2
Trachea
Macrophage2

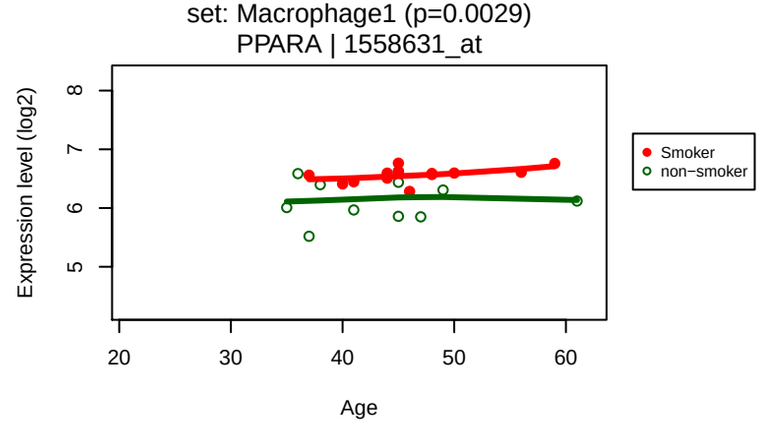

set: SmallAirway1 (p=0.0296)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker

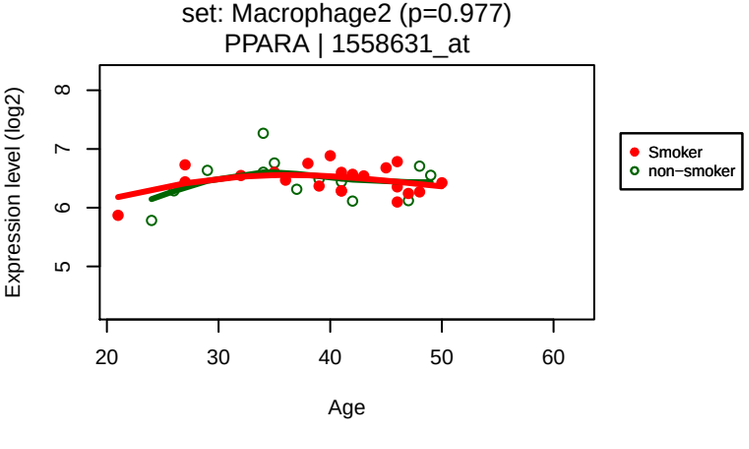

set: SmallAirway2 (p=0.2703)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker

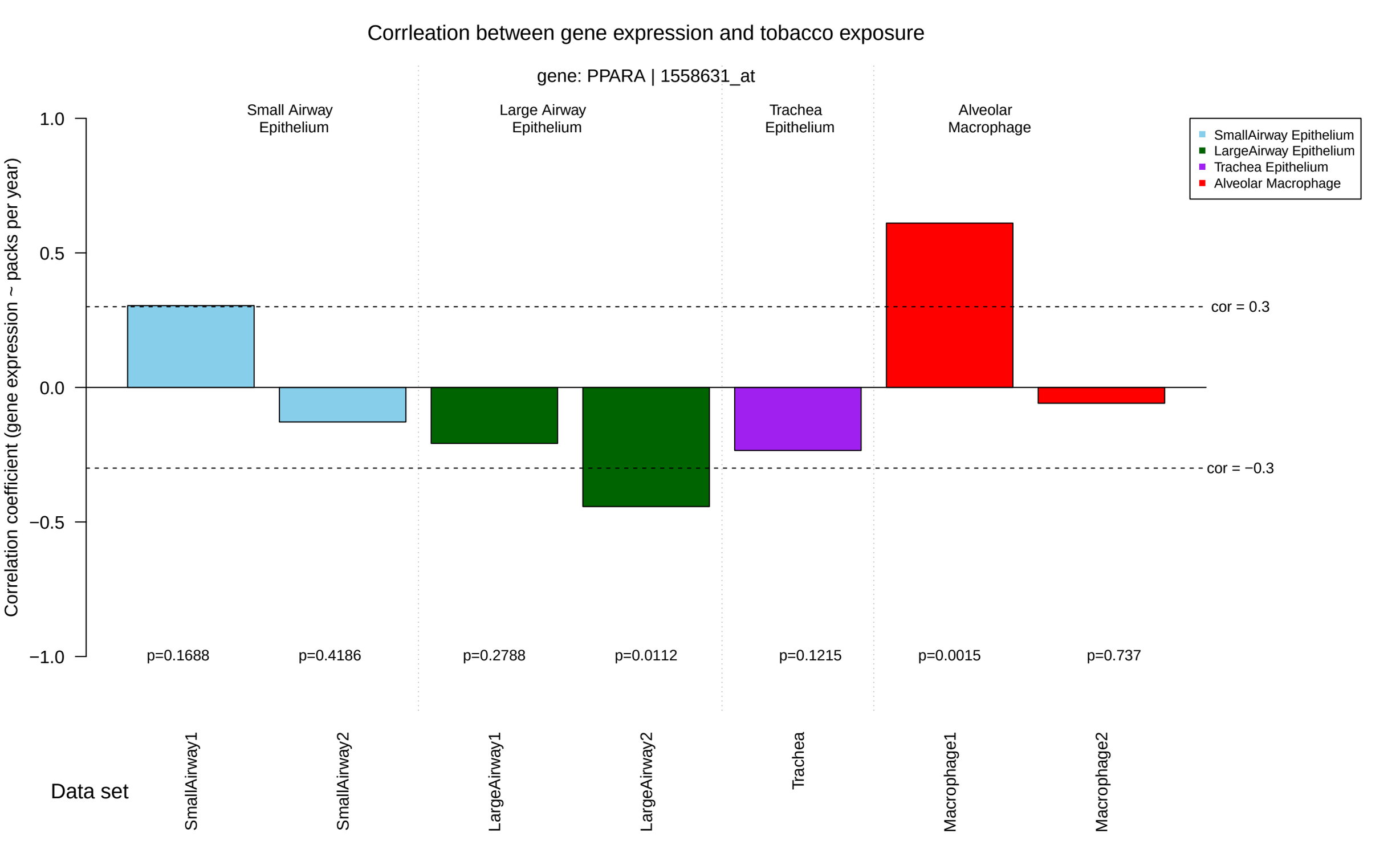

set: LargeAirway1 (p=0.1345)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1167)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3324)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0029)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.977)
PPARA | 1558631_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPARA | 1558631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1688
p=0.4186
p=0.2788
p=0.0112
p=0.1215
p=0.0015
p=0.737
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2